

Farzaneh Fayazbakhsh

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Personal Statement

Interdisciplinary researcher with a foundation in bioinformatics, cancer biology and molecular biotechnology, combining Python/R analysis with experimental work and scientific communication. My experience includes tumor models, microscopy, sequencing data, reproducible workflows and multimodal transcriptomics; I am motivated to deepen my expertise in machine learning and medical image analysis for clinically useful precision-oncology tools.

Education

M.Sc. Swedish University of Agricultural Sciences , Bioinformatics	2025–present
- Highest grade (5/5) in Bioinformatics course - Thesis: Exploring the Potential of Fragile Sites as Markers for Lineage Tracing	
KTH Royal Institute of Technology , Single-Cell & Spatial Transcriptomics Data Analysis Course	2024
M.Sc. Tehran University of Medical Sciences , Medical Nanotechnology	Iran
- Awarded as the Top Master's Graduate (GPA: 4/4) - Thesis: Evaluating the Antioxidant Potential of Resveratrol-Gold Nanoparticles in Preventing Oxidative Stress in Endothelium on a Chip	2017–2022
B.Sc. Alzahra University , Cellular and Molecular Biology (Biotechnology Specialization)	Iran
	2012–2016

Bioinformatics & Computational Biology Projects

Automated Raw Sequencing Data Quality Control Pipeline

- Built an automated, parallelized Bash orchestration workflow designed for execution on High-Performance Computing (HPC) clusters via SLURM scheduling.
- Deployed FastQC and fastp dynamically inside isolated, version-controlled Singularity and Apptainer containers to automate raw sequence pairing, quality checking, and adapter trimming.
- Implemented multi-core processor allocation inside container runtimes to optimize throughput when executing string substitutions and batch file calculations.
- Orchestrated MultiQC container tasks to aggregate and output comprehensive data quality matrices, tracking processing adjustments across all sample cohorts simultaneously.

Tumor Microenvironment Profiling: scRNA-Seq & Spatial Transcriptomics

- Engineered end-to-end transcriptomic pipelines in R using Seurat and SCTransform to map high-dimensional gene expression across breast cancer tissue cohorts.
- Executed data pre-processing, Quality Control (QC), linear (PCA), and non-linear (t-SNE / UMAP) dimensionality reduction alongside graph-based clustering to resolve distinct cell populations and structural tissue features.
- Containerized both analysis environments using Docker and Docker Compose, ensuring full cross-platform compatibility and seamless runtime execution across varied infrastructure platforms.
- Managed pipeline documentation, software repositories, and strict workflow reproducibility utilizing Git and GitHub version control standards.

Computational Biology & Bioinformatics Skills

Programming & Pipeline Automation: Python, R, UNIX/Linux Shell, Nextflow (pipeline development), Git, and GitHub for version control

Genomics, High-Throughput Sequencing & Genetics: Practical experience in NGS data analysis, Whole Genome Sequencing (WGS), RNA-Seq processing, GWAS, QTL analysis, gene mapping, structural variation, metagenomics, genome annotation, and phylogenomics (MEGA, Clustal Omega)

Structural Biology, Modeling & Machine Learning: Sequence analysis, primer design, AlphaFold (protein structure prediction), PyMOL (molecular visualization), and foundational familiarity with deep learning frameworks

Infrastructure, HPC & DevOps: Hands-on experience with High-Performance Computing (HPC) clusters, containerization (Docker, Docker Compose, Apptainer/Singularity), and implementation of FAIR Research Data Principles

Research Experience

- Karolinska Institute & SciLifeLab**, Master's Thesis Research 2026–present
- Investigating replication-stress-induced genomic instability at common fragile sites using targeted long-read sequencing.
 - Performing neuroblastoma cell culture, drug treatments, DNA extraction, and long-range PCR.
 - Developed bioinformatics workflows using Python, Linux and HPC environments for long-read sequencing analysis.
 - Performed structural variant discovery, breakpoint characterization and endogenous barcode identification.
- KTH Royal Institute of Technology & SciLifeLab**, Research Engineer Sweden 2023–2024
- Independently developed Tumor-on-a-Chip models for in vitro anti-cancer drug screening.
 - Optimized layer-by-layer cellulose nanofibril coatings to enhance cancer spheroid formation.
 - Contributed to projects on Brain-on-a-Chip models for metabolic studies.
 - Functionalized electrochemical sensors for ketone detection.
- Tehran University of Medical Sciences**, Master's Thesis Research Iran 2018–2022
- Synthesized gold nanoparticles conjugated with polyphenols.
 - Developed microfluidic devices for endothelial cell culture.
 - Modeled early-stage endothelial dysfunction and conducted oxidative stress assays.
 - Assisted in preparing a PET/PDMS/Collagen nanofibrous membrane for microfluidic cell culture.
- Gen-Iran Lab**, Pathology and Cytology Intern Tehran, Iran 2023
- Hands-on experience in tissue preparation, fixation, embedding, and sectioning.
 - Performed histological and immunohistochemical staining.

Peer-Reviewed Publications

- Evaluating the Antioxidant Potential of Resveratrol-Gold Nanoparticles in Preventing Oxidative Stress in Endothelium on a Chip** 2023
Farzaneh Fayazbakhsh, Fatemeh Hataminia, Houra Mobaleghol Eslam, Mohammad Ajoudanian, Sharmin Kharrazi, Kazem Sharifi, Hossein Ghanbari
Scientific Reports
- Metabolic Assessment of iPSC-Derived Neurons Under Ketogenic and Normal Diets: Ketone Sensor Development and Comparative Analysis of Neuronal Metabolism** 2026
Rohollah Nasiri*, **Farzaneh Fayazbakhsh***, Reyhaneh Sanei, Tingting Wu, Nayere Taebnia, Rouhollah Habibbey, Omid Fotouhi, John Cognetti, Volker M. Lauschke, Aman Russom, Anna Herland
CellPress
- Assessing Layer-by-Layer Assembly of Cellulose Nanofibrils in Pancreatic Tumor Spheroid Formation** 2023
Negar Abbasi Aval, Ekeram Lahchaichi, Oana Tudoran, **Farzaneh Fayazbakhsh**, Rainer Heuchel, Matthias Löhr, Torbjörn Pettersson, Aman Russom
Biomedicines
- Characterization and Numerical Simulation of a New Microfluidic Device for Cell Investigations** 2024
H Meslam, F Hataminia, H Asadi-Saghandi, **F Fayazbakhsh**, N Tabatabaei, Hossein Ghanbari
Frontiers in Lab on a Chip Technologies
- Nanofibers in Respiratory Masks: An Alternative to Prevent Pathogen Transmission** 2022
Tabatabaei SN, Faridi-Majidi R, Boroumand S, Norouz F, Rahmani M, Rezaie F, **Fayazbakhsh F**, Faridi-Majidi R
IEEE Transactions on NanoBioscience

Wet Lab & Experimental Expertise

- Cell Culture & Cancer Biology:** 2D & 3D cell culture, tumor spheroid formation and organoid culture, in vitro cancer biology drug screening
- Molecular Biology & Genomics Workflows:** DNA extraction, downstream cellular and molecular assessments, PCR / Real-Time PCR, agarose gel electrophoresis, Agilent TapeStation analysis, flow cytometry, fluorescent microscopy

Microfluidics & Biosensors: Organ-on-a-Chip modeling, PDMS-based microfluidic systems fabrication, surface and electrochemical sensor functionalization

Histology & Pathology: Histological sample preparation, H&E staining, immunohistochemistry (IHC)

Nanotechnology & Nanomaterial Synthesis: Green synthesis of gold nanoparticles, nanofiber electrospinning, cellulose nanofibrils (CNF) coating optimization

Conference Presentations

Spheroid Formation in Cellulose Nanofibrils-Coated Microfluidic Chips

Poster presentation (first author)

μTAS 2024, Montreal, Canada
2024

Design of a Microfluidic Electrochemical Ketone Sensor

Oral presentation (co-author)

Swedish Microfluidics in Life Science, KTH, Sweden
2023

Investigation of Effect of Different Diets on Brain Energy Metabolism Using Organ-on-a-Chip

Poster presentation (co-author)

MPS World Summit, Berlin, Germany
2023

Preparation of PET/PDMS/Collagen Nanofibrous Membrane Embedded in a Microfluidic Device for Organ-on-Chip Applications

Poster presentation (co-author)

The 3rd Conference on Nanofibers, Tehran University of Medical Sciences
2022

Professional & Personal Skills

Bridging Wet & Dry Lab: Confident working in both experimental labs and computational environments, making it easy to translate biological questions into data-driven answers

Independent Problem Solver: Highly autonomous in research environments, with a proven ability to troubleshoot bottlenecks and quickly teach myself new tools to keep projects moving forward

Collaborative Team Player: Experienced in international, interdisciplinary teams such as KTH, Karolinska Institute and SciLifeLab, with a strong focus on knowledge-sharing and helping colleagues

Clear Communicator & Fast Learner: Skilled at explaining complex data clearly through peer-reviewed publications and conference presentations; highly adaptable and quick to pick up new scientific topics and skills

Teaching & Mentorship

KTH Royal Institute of Technology, Intern Supervisor

Sweden

- Guided an intern in laboratory techniques.

Volunteer Service, Science Teacher

- Taught science to underprivileged children.

Awards & Honors

Top Master's Graduate

Tehran University of Medical Sciences
2023

LUSH Prize Candidate

2022

Shortlisted for the Young Researcher Award.

Darolfonoon Student Award

2020

For voluntary assistance in nanofiber mask production during the COVID-19 pandemic.

Top Article Award

Prize winner.

First National
and Student
Conference on
Applied Biology,
Alzahra Univer-
sity, Iran
2014

Top 5% in Iran's Konkour for Master's Degree University Entrance

Iran

Ranked 37th in Iran's national Konkour for the Master's Degree University Entrance Exam.

2016

References

Dr. Kasper Karlsson #linebreak() Department of Oncology-Pathology, Karolinska Institute #linebreak() kasper.karlsson@ki.se



Dr. Erik Bongcam Rudloff #linebreak() Department of Animal Biosciences, Swedish University of Agricultural Sciences #linebreak() erik.bongcam@slu.se ▼

Dr. Aman Russom #linebreak() Nanobiotechnology Division, Protein Science Department, KTH Royal Institute of Technology #linebreak() aman.russom@scilifelab.se ▼

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Dr. Rohollah Nasiri #linebreak() Radiation Oncology Department, Stanford University #linebreak() rhnasiri@stanford.edu ▼